SECTION 8.D
Divide Polynomials

Polynomial division extends the structure of numerical long division into algebra. Just as numbers can be divided to produce a quotient and remainder, polynomials can also be divided systematically. In this section, you will learn multiple methods for dividing polynomials and interpreting the remainder.

Learning Objectives

- I can divide polynomials using long division, connecting the process to numerical long division and interpreting remainders.
- I can apply synthetic division as an efficient algorithm for dividing by linear binomials.
- I can use the polynomial division box method to organize and confirm division results.

Introduction: Extending Division to Algebra

Polynomial division follows the same structural principles as arithmetic division. You determine how many times the divisor fits into the leading term of the dividend, multiply, subtract, and repeat.

Division may result in:

- An exact quotient (no remainder)
- A remainder polynomial

When a remainder exists, it can be written as:

$$\frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)}$$

Real-World Connection

Engineers use polynomial division when modeling systems involving ratios of polynomials, especially in control systems and signal processing. Division reveals hidden structure and simplifies models for analysis.

Historical Note

Polynomial division techniques were formalized in the 17th century as algebra matured into symbolic form. Synthetic division later emerged as a streamlined algorithm for linear divisors, reducing repetitive computation.

Key Vocabulary and Concepts

Dividend: The polynomial being divided.

Divisor: The polynomial you divide by.

Quotient: The result of division.

Remainder: The polynomial left after division.

Synthetic Division: A shortcut method for dividing by linear binomials of the form $x - c$.

Division Algorithm:

$$P(x) = D(x)Q(x) + R(x)$$

Core Methods

There are three primary methods for dividing polynomials:

1. Long Division
2. Box Method
3. Synthetic Division

Example 1 — Long Division

Divide:

$$\frac{x^3 - 2x^2 + 4x - 8}{x - 2}$$

Example 2 — Box Method

Divide:

$$\frac{2x^3 + 3x^2 - 5x + 6}{x + 2}$$

Example 3 — Synthetic Division

Divide:

$$\frac{x^4 - 3x^3 + 2x^2 - x + 5}{x - 1}$$

Summary of Key Concepts

- Polynomial division follows the same structural steps as numerical long division.
- The division algorithm states: $P(x) = D(x)Q(x) + R(x)$.
- Synthetic division applies only when dividing by linear binomials of the form $x - c$.
- The box method provides visual structure and organization.
- A remainder can be written as a fractional term over the original divisor.

Division Strategy Checklist

1. Rewrite the dividend in descending powers.
2. Include zero placeholders for missing terms.
3. Divide the leading terms.
4. Multiply the divisor by the new term.
5. Subtract and repeat.
6. Write the remainder properly.

Common Mistakes

Forgetting Missing Terms

When a polynomial skips a power (for example, no x^2 term), a zero placeholder must be included before division.

Sign Errors During Subtraction

Subtraction errors are common when distributing negative signs across polynomials.

Using Synthetic Division Incorrectly

Synthetic division only works for divisors of the form $x - c$. It does not apply to quadratic or higher-degree divisors.

Dropping the Remainder

If division is not exact, the remainder must be written as a fraction over the divisor.

Practice Problems

Easy

1. Divide:

$$\frac{x^2 + 5x + 6}{x + 2}$$

2. Divide:

$$\frac{x^3 - 4x}{x}$$

3. Divide:

$$\frac{x^3 + 3x^2 - 4}{x - 1}$$

Medium

4. Divide:

$$\frac{2x^3 + 7x^2 - x - 6}{x + 3}$$

5. Divide:

$$\frac{x^4 - 5x^2 + 4}{x^2 - 1}$$

6. Use synthetic division:

$$\frac{x^4 - 3x^3 + 2x - 5}{x - 2}$$

Hard

7. Divide:

$$\frac{3x^4 - 5x^3 + x^2 - 4x + 6}{x - 1}$$

8. Divide:

$$\frac{2x^3 - x^2 + 4x - 3}{x + 2}$$

9. Divide:

$$\frac{x^5 - x^3 + 2x - 1}{x - 1}$$

Challenge

10. Use the box method to divide:

$$\frac{x^3 + 4x^2 - 5x - 20}{x + 4}$$

11. Create a polynomial that leaves a remainder of 3 when divided by $x - 2$.

12. Explain how the Remainder Theorem relates to synthetic division.

Justify

13. Justify why synthetic division works only for linear divisors.

14. Explain why including zero placeholders is necessary during polynomial division.

15. Justify why the remainder must be written over the original divisor.

16. Explain how polynomial division connects to rational functions.

17. Explain how the Division Algorithm applies to higher-degree polynomials.

18. Justify why polynomial division preserves equality in rational expressions.

Solutions

Easy

1.

$$\frac{x^2 + 5x + 6}{x + 2} = \frac{(x + 2)(x + 3)}{x + 2} = x + 3 \quad (x \neq -2)$$

2.

$$\frac{x^3 - 4x}{x} = \frac{x(x^2 - 4)}{x} = x^2 - 4 \quad (x \neq 0)$$

3. Synthetic division yields:

$$x^2 + 4x + 4$$

Medium

4. Long division gives:

$$2x^2 + x - 4$$

5.

$$\frac{x^4 - 5x^2 + 4}{x^2 - 1} = \frac{(x^2 - 4)(x^2 - 1)}{x^2 - 1} = x^2 - 4 \quad (x \neq \pm 1)$$

6. Synthetic division yields:

$$x^3 - x^2 - 2x - 2$$

with remainder:

$$-9$$

Hard

7. Long division yields:

$$3x^3 - 2x^2 - x - 5$$

with remainder:

$$1$$

8. Box method yields:

$$2x^2 - 5x + 14$$

with remainder:

$$-31$$

9. Synthetic division yields:

$$x^4 + x^3 + 0x^2 + 2$$

with remainder:

$$1$$

Challenge

10. Box method yields:

$$x^2 - x - 5$$

11. Example:

$$(x - 2)(x^2 + 1) + 3$$

12. The Remainder Theorem states that evaluating $P(c)$ gives the remainder when dividing by $x - c$. Synthetic division performs this efficiently.

Justify

13. Synthetic division relies on the divisor being linear in the form $x - c$. Higher-degree divisors require long division or box method.

14. Zero placeholders maintain correct term alignment during subtraction in the division process.

15. Writing the remainder over the original divisor preserves equality according to the Division Algorithm.

16. Polynomial division allows rational functions to be rewritten as a polynomial plus a proper rational remainder.

17. The Division Algorithm applies to all polynomial degrees because it is based on distributive structure.

18. Division preserves equality because each subtraction step maintains equivalent expressions.